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# Navigating the Human–Robot Interface—Exploring Human Interactions and Perceptions with Social and Telepresence Robots

**Eva Mårell-Olsson [1,*], Suna Bensch [2], Thomas Hellström [2], Hannah Alm [3], Amanda Hyllbrant [3], Mimmi Leonardson [3] and Sanna Westberg [3]**

[1] Department of Education, Umeå University, 901 87 Umeå, Sweden
[2] Department of Computing Sciences, Umeå University, 901 87 Umeå, Sweden; suna.bensch@umu.se (S.B.); thomas.hellstrom@umu.se (T.H.)
[3] Department of Mathematics and Mathematical Statistics, Umeå University, 901 87 Umeå, Sweden
* Correspondence: eva.marell-olsson@umu.se

**Abstract:** This study investigates user experiences of interactions with two types of robots: Pepper, a social humanoid robot, and Double 3, a self-driving telepresence robot. Conducted in a controlled setting with a specific participant group, this research aims to understand how the design and functionality of these robots influence user perception, interaction patterns, and emotional responses. The findings reveal diverse participant reactions, highlighting the importance of adaptability, effective communication, autonomy, and perceived credibility in robot design. Participants showed mixed responses to human-like emotional displays and expressed a desire for robots capable of more nuanced and reliable behaviors. Trust in robots was influenced by their perceived functionality and reliability. Despite limitations in sample size, the study provides insights into the ethical and social considerations of integrating AI in public and professional spaces, offering guidance for enhancing user-centered designs and expanding applications for social and telepresence robots in society.

**Keywords:** human-robot interaction (HRI); social and telepresence robots; user experience; Pepper robot; Double 3 robot

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Academic Editors: Yutaka Ishibashi, Tomislav Jagušt, Peter Seow Sen Kee, Martina Holenko Dlab and Ana Sović Kržić

Received: 6 November 2024
Revised: 9 January 2025
Accepted: 22 January 2025
Published: 23 January 2025

**Citation:** Mårell-Olsson, E.; Bensch, S.; Hellström, T.; Alm, H.; Hyllbrant, A.; Leonardson, M.; Westberg, S. Navigating the Human–Robot Interface—Exploring Human Interactions and Perceptions with Social and Telepresence Robots. *Appl. Sci.* **2025**, *15*, 1127. https://doi.org/10.3390/app15031127

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## 1. Introduction

The rapid advancement of robotics and artificial intelligence (AI) has transformed how humans perceive and engage with intelligent technologies. A key aspect of this evolution is the design of robots that function not only as machines but as social partners, tailored to coexist and collaborate with humans in diverse personal and professional environments. Recent studies emphasize that human–robot interaction (HRI) has emerged as a significant area of research, focusing on how different aspects of robot design—including embodiment, autonomy, and communication styles—influence user perception, trust, and engagement [1–3].

Embodiment, or the physical form of a robot, is particularly critical in shaping user experience. Previous studies have demonstrated that physical embodiment can significantly affect perceptions of believability, emotional engagement, and usability [4]. More recent research has reinforced these findings. Zhu et al. [1] highlighted that a robot's physical appearance, movements, and communication styles can influence user interaction, with human-like gestures leading to greater trust and satisfaction. Similarly, Rossi et al. [2]

emphasized the importance of coherent nonverbal communication, noting that robots displaying synchronized gestures and physical presence significantly enhanced users' emotional responses. Furthermore, Sorrentino et al. [3] discussed how robots' abilities to convey emotions through embodiment are crucial for fostering natural and empathetic interactions. Carreno-Medrano et al. [5] also highlighted how aligning robot behaviors with user expectations fosters more meaningful and effective interactions, particularly in collaborative settings. These studies underscore that embodiment directly impacts the quality of interaction, emotional engagement, and user satisfaction.

Additionally, Sheridan [6] noted that specialized robot designs tailored to specific tasks tend to enhance both usability and user satisfaction. Mutlu [7] further observed that users often respond differently to robots with varying degrees of human likeness, suggesting that embodiment plays a key role in shaping perceptions of credibility and believability.

The integration of robots into domains such as healthcare, education, and retail underscore the importance of understanding HRI. For instance, D'Alfonso et al. [8] found that users are more likely to share sensitive information with robots perceived as neutral entities, emphasizing the potential role of social robots in therapy and counseling. Recent studies further highlight the growing applications of robots in these domains. In healthcare, Chen et al. [9] discussed how robotic assistants are increasingly being used for elder care, supporting both physical and emotional needs. Ramirez and Carter [10] demonstrated that generative AI-driven robots improve customer service through personalized recommendations, enhancing engagement and satisfaction in retail settings. Similarly, Kim and Lee [11] found that social robots in STEM education significantly enhance interactive learning experiences, particularly in collaborative classroom environments. Su et al. [12] further elaborated on multimodal human–robot interaction, showing how robots capable of processing speech and gestures create more engaging educational settings.

Despite the growing body of research, significant gaps remain in understanding how the interplay of physical embodiment and functionality influences user interaction. While studies have examined humanoid robots like Pepper in customer service settings [13,14] and telepresence robots like Double 3 in remote workplaces [15,16], limited attention has been given to direct comparisons between these distinct embodiments. Furthermore, as Wirtz et al. [17] noted, the success of social and telepresence robots hinges on their ability to adapt to diverse user needs, yet few studies have systematically examined how these robots perform in controlled, comparative experiments.

This study investigates user experiences with two distinct types of robots: Pepper, a social humanoid robot, and Double 3, a self-driving telepresence robot. Specifically, the research focuses on understanding how differences in the design and functionality of these robots influence user perceptions, interaction patterns, and emotional responses. By exploring these aspects, the study seeks to provide insights into the role of robot design and functionality in shaping user experiences during human–robot interactions.

Building on foundational studies [7,18], this research also seeks to understand the contextual and design factors that impact user experience. In doing so, we aim to offer practical recommendations for enhancing robot design and expanding their applications in public and professional settings.

By investigating the nuances of user experiences, this study contributes to the broader understanding of how robots can be effectively integrated into everyday life. The findings have the potential to inform the development of more adaptable, interactive, and credible robots, paving the way for their seamless integration into society.

The following sections of this paper are structured to provide a comprehensive analysis of the study. The methodology outlines the experimental setup and participant recruitment process, emphasizing the qualitative approach. The results highlight participants' percep-

tions and emotional responses, offering insights into the strengths and limitations of each robot. The discussion contextualizes these findings within the existing literature, drawing connections to broader implications for AI design and ethical considerations. Finally, the conclusions synthesize the main contributions of the study and propose directions for future research, focusing on enhancing user-centered design and expanding applications for robots in various societal contexts.

*Background*

The study of human interaction with artificial agents has become increasingly important for understanding the nuances of Human-Robot Interaction (HRI). Central to this field is the concept of "embodiment" in artificial agents, which significantly influences user perception and engagement. Cassell et al. [4] noted that the physical manifestation of these agents plays a crucial role in shaping how humans interact with and perceive them. Over the past decade, extensive research has focused on categorizing and understanding these various embodiments. Mutlu [7] offers valuable insights into this area, showing how people's experiences and cognitive responses vary between virtual and physical embodiments of interactive agents. Beyond the type of embodiment, its degree and design collectively shape human perception and interaction dynamics.

Furthering this discourse, D'Alfonso et al. [8] suggest that people may find it easier to share private or sensitive information with a robot rather than a human, due to the robot's perceived neutrality. This insight raises important questions about the potential roles of robots in settings where confidentiality or discretion is essential.

In recent years, the integration of robotics across diverse sectors has marked a significant evolution in our relationship with technology. Sheridan [6] emphasizes the transformative impact of this integration on HRI, noting that it has facilitated substantial advancements in both technology and interaction models. Siciliano et al. [18] further elaborate on this shift, discussing the rise of specialized robots designed for specific tasks and environments. This specialization is reflected in the development of social humanoid robots like Pepper [13] and telepresence robots like Double 3 [19]. These robots represent a paradigm shift in creating interactive, user-centered solutions that seamlessly blend digital and physical realms, as envisioned by Breazeal [20].

Pepper and Double 3 stand out for their capabilities and applications, serving as examples of recent technological progress. Pepper, with its humanoid appearance and interactive features, has been adopted in customer service [17], healthcare [21], and education [22]. Similarly, Double 3, equipped with telepresence capabilities, is transforming remote workplace interactions [23], enabling new possibilities for bridging geographical distances.

The primary aim of this study is to investigate how users interact with two distinct types of social and telepresence robots—Pepper, a social humanoid robot, and Double 3, a telepresence robot. Furthermore, the research aims to understand how the design and functionality of these robots influence user perception, interaction patterns, and emotional responses. Additionally, the study seeks to identify key factors that contribute to user experience and satisfaction with these robots, considering their specific technological capabilities and limitations. The study also examines participants' perspectives on integrating such robots into everyday environments.

The central research question guiding this study is: How do users perceive and interact with these social and telepresence robots, and what are the implications of these interactions for user experience, technological expectations, and social dynamics?

The specific research questions explored in this study are:

1. What common experiences and reactions do participants have when interacting with each type of robot?
2. What unique experiences and reactions do participants exhibit when interacting with each type of robot?
3. What specific challenges or areas for improvement can be identified to enhance the user experience with these social and telepresence robots?

The following sections will describe the Pepper and Double 3 robots in detail, focusing on their design, functionality, and impact in diverse interaction settings.

## 2. Materials and Methods

### 2.1. The Pepper Robot

The Pepper robot (see Figure 1), developed by the French company Aldebaran (Lyon, France) [24], is a humanoid robot with a human-like appearance and behavior. The 120 cm tall robot has various sensors, enabling it to navigate safely. It can detect and recognize faces (although the software is a bit outdated and the performance is not quite state-of-the-art), and it interacts through a touchscreen and voice communication system. The robot's articulated speech functionality synchronizes gestures and body posture to align naturally with verbal communication.



**Figure 1.** A Pepper robot interacting with one of the participants in the test.

In healthcare settings, Pepper has been proposed to support staff and engage patients. Research by Alemi et al. [25] highlights Pepper's potential for increasing patient engagement and aiding in care settings. Pepper has also been deployed in elder care, assisting with cognitive exercises and providing companionship [13]. While these applications suggest potential benefits for geriatric care, further studies are needed to assess Pepper's effectiveness and suitability in these application areas. In educational contexts, studies by Tanaka and Matsuzoe [14] suggest that Pepper may enhance learning experiences, particularly in programming and robotics education. In the retail and service sectors, the Pepper robot is suggested to improve customer satisfaction [17].

In summary, Pepper shows potential in multiple fields, including healthcare, elder care, education, and retail.

### 2.2. Double 3 Robot

Double 3, developed by Double Robotics (Redwood City, CA, USA), is a telepresence robot that allows remote control by an operator via a standard computer. The operator can view the robot's environment through a camera mounted on the robot, while their face is displayed on the robot's screen (see Figure 2). Double 3 is equipped with an audio system that enables the operator to hear and speak through the robot's loudspeakers, facilitating two-way communication. Its two-wheeled, self-balancing design contributes to stability

and ease of navigation. The robot's 3D sensors support navigation and obstacle avoidance, allowing the operator to perceive a physical presence in various settings such as offices, educational institutions, and healthcare facilities [19].



**Figure 2.** A remote user interacting through the Double 3 robot with one of the participants in the test.

The Double 3 has been tested in several different contexts. In education, it has been utilized to facilitate remote classroom participation, enabling students who cannot be physically present for learning activities. Research by Zhao and Okamoto [15] indicates that Double 3 can enhance the learning experience by providing accessible educational opportunities, though its effectiveness may vary based on classroom dynamics and student needs.

In corporate settings, Double 3 has been applied for remote work and teleconferencing. Studies by Neustaedter et al. [16] highlight how it may improve communication and collaboration within distributed teams, although its impact varies with the workplace contexts and the team dynamics.

In healthcare, Double 3 has been used for remote patient consultations. As noted by Smith [26], this application represents a novel approach to telemedicine, improving access to medical consultations for patients in remote or underserved areas. However, the appropriateness and efficacy of such applications depend on the specific medical context.

Overall, existing research presents Double 3 as a versatile tool for enhancing remote interaction in several sectors. However, the degree to which the robot can replace physical presence remains an area for further exploration.

### *2.3. Theoretical Framework*

The theoretical framework of this study is based on five mechanisms that represent constructs in which distinct differences can be observed in human experiences with various robot representations: situativity, interactivity, agency, proxemics, and believability [7]. These mechanisms collectively contribute to understanding the dynamics of human–robot interaction, highlighting key dimensions in how users perceive and engage with robots.

Situativity in robot-user interactions refers to how context influences a robot's behavior. This mechanism examines the degree to which a robot, designed for specific applications, can adapt to the user's environment. For instance, users might repurpose a robot for varied scenarios, affecting engagement and perceived realism. Unlike virtual agents limited to scripted settings, physical robots interact dynamically within user-defined physical spaces [7]. Both robots in this study are situated in the user's physical environment, though each is designed with different purposes. Differences in situativity impact how users perceive and interact with the robots; for example, one robot may be more suited for tasks requiring physical handling of objects, while the other may excel in tasks that involve mobility in different settings.

Interactivity refers to the extent to which users can engage with the robot [7]. This mechanism focuses on the structure and evolution of interactions, both in physical and vir-

tual environments. Interactivity influences user satisfaction and the perceived intelligence of a robot. While interactions with virtual agents are often scripted and limited, physical robots allow for more user-directed engagement. In this study, both physical robots enable natural and intuitive interactions, though the specific capabilities of each vary. For instance, one robot may be better suited for tasks requiring physical assistance, while the other may excel in information retrieval.

Agency denotes the degree to which a robot can act autonomously and influence its surroundings, as well as its perception as an independent entity with its own goals [7]. Agency affects users' trust and perceptions of the robot's competence. Physical robots are often seen as more flexible in acting independently due to their ability to impact the physical environment, while virtual agents are constrained by their lack of physical autonomy. In this study, both robots have physical forms and can move autonomously, though their specific behaviors differ. For example, one robot may be suited to tasks that require autonomous decision-making, while the other may be designed to follow a structured script.

Proxemics refers to the spatial relationship between the user and the robot, addressing how a robot's proximity and behavior toward the user affect interaction and user behavior [7]. Proxemics influence user comfort and the perceived social presence of the robot. Physical robots can adjust proximity dynamically, while virtual agents are restricted by their virtual settings. In this study, both robots adhere to social norms for spatial relations, though each has unique capabilities. For example, one robot may be more appropriate for tasks that require close proximity, while the other may be designed for interaction at a distance.

Believability refers to the degree to which a robot's appearance and behavior are perceived as realistic and convincing [7]. Believability affects user engagement and perceived social presence. Physical robots are often seen as more believable because of their tangible nature and ability to physically interact with users, unlike virtual agents. Both robots in this study, by virtue of their physical forms, can interact with users in natural ways, although design and functionality differences may affect perceived believability. For example, one robot may have a human-like appearance and behavior, while the other may adopt a more mechanical design.

### *2.4. Experimental Setting*

The study was conducted in a simulated warehouse environment designed to resemble real-world working conditions. The warehouse included shelves, aisles, and clearly marked areas to facilitate the navigation of both robots and participants. Efforts were made to minimize distractions, and all interactions took place in a quiet and controlled setting. Participants interacted with the robots one at a time, ensuring consistent and comparable experiences.

The Pepper and Double 3 robots were positioned at designated starting points at the beginning of each session. The remote operator navigated the Double 3 robot manually using a computer interface, while Pepper's actions were pre-programmed and activated through a Wizard-of-Oz (WoZ) system [27]. Participants received verbal instructions and physical guidance from the robots as they completed tasks within the simulated warehouse. All participants interacted with the robots in the same order, starting with the Double 3 robot followed by the Pepper robot. This consistent order ensured uniformity in the interaction sequence and allowed for a straightforward comparison of participant responses to the two robots.

Before each session, participants were briefed individually on the procedure and encouraged to interact naturally with the robots. The experimental team was present

to monitor and ensure the smooth operation of the robots but did not intervene during the interaction unless necessary. This setup aimed to balance ecological validity with experimental control, providing insights into real-world interaction dynamics.

The participants, aged between 23 and 30 years, included 8 men and 2 women. Participants were recruited through invitations sent to university students, primarily targeting those in industrial engineering and construction engineering programs. Of the 10 respondents, six were enrolled in the construction engineering program, and four were studying civil engineering within industrial economics and management. While this participant group provides valuable insights into human–robot interaction, we acknowledge that the sample is not representative due to its limited scope, both in terms of academic backgrounds and gender distribution. To maximize data collection, all interested individuals were invited to participate.

### 2.5. Design-Based Research Methods

This study employed a design-based research approach, aligning with the goal of exploring user interactions with robots. Design-based research is known for its iterative, adaptive, and collaborative nature, merging empirical investigation with theory-driven design in interactive environments. This approach is particularly suited for studying human–robot interactions, as it aims to enhance user experience and develop context-specific design principles [28,29].

In this study, the design-based approach provided a structured yet flexible framework for examining user experiences with the Double 3 and Pepper robots. Through close participant engagement, the research team could develop, test, and refine interaction methods that addressed specific user interaction challenges, thus contributing to the broader field of human–robot interaction.

The method followed several key steps: identifying the problem, collaborating with stakeholders, designing the intervention (i.e., robot interaction setup), implementing and testing the intervention, analyzing the data, refining the approach, and documenting and sharing findings [29]. Initially, the study identified a core research question—understanding user experiences and interactions with different types of robots. The interaction setup was then collaboratively designed and tailored to the distinct functionalities of each robot.

The study included multiple rounds of interaction sessions with each participant interacting with the robots once, followed by detailed interviews and observations. Data from each session were analyzed to refine the interaction setup for the next round. Researchers were present during each session to facilitate the process and gain first-hand insights into participants' reactions. After each round, findings were reviewed, and adjustments were made, enabling a more nuanced understanding of user experiences with the robots.

The iterative process played a critical role in shaping the interaction design. After each session, feedback and observations were incorporated, allowing adjustments to both the interaction setup and the interview guide. As the study progressed, participants' initial skepticism and hesitation toward the robots became a focal area of investigation. To further understand these emotional and psychological aspects, the interview questions were expanded to include items related to the robots' behavior and their perceived ability to convey human emotions. This addition enabled the collection of more in-depth data, revealing a complex picture of participants' emotional responses to the robots.

During the interviews, participants shared insights on the emotional challenges they faced while interacting with the robots. These findings highlight areas for potential technological improvements aimed at reducing user uncertainty and discomfort. By addressing such factors, this study provides valuable insights into refining robot design and improving user experience in human–robot interactions.

### 2.6. Data Collection

Observations. A set of predefined observation points, including situations where participants interacted with specific robot functions or exhibited emotional responses toward the robots, were established. By systematically examining and analyzing these key points, detailed data on the interactions were collected. Data were gathered through both manual observation and video recording for later review, allowing for in-depth analysis of participant-robot interactions.

Interviews. Semi-structured interviews were conducted with participants to capture their reflections and experiences interacting with the robots. To maintain consistency and relevance, a semi-structured interview guide was developed, grounded in the study's purpose, research questions, and theoretical framework. Questions were designed to encourage open-ended responses, such as the following: "What was your overall experience with each robot?", "How did meeting these two different robots—one with and one without a human face—differ?", "Do you see potential uses for these types of robots in companies or workplaces?", "To what extent do you think these robots could replace a human in similar scenarios?", "Was there anything specific about the robots' behavior that stood out to you?", and "How did you perceive the robots' ability to convey human emotions, like when the Pepper robot appeared happy or sad?"

The interview guide also included questions about participants' views on the future of robots, such as the following: "How do you envision the future use of robots? Do you think they will become more common, and if so, why?", "Can you provide examples of areas where you think robots will be used?", "What challenges do you foresee in the integration of robots? Can you give specific examples?", and "How do you personally feel about a future where robots are more prevalent?"

All interviews were conducted face-to-face, allowing participants to engage directly with researchers. Each interview lasted approximately 15–20 min, and all sessions were audio recorded and transcribed verbatim.

### 2.7. Exploratory Analysis

To interpret and analyze the empirical data collected in this user study and identify key themes and emerging patterns, we employed thematic analysis [30]. Thematic analysis is a central method in qualitative research for coding and interpreting data, helping researchers uncover meaningful insights. As described by Boyatzis [31], this process involves two perspectives: "seeing" and "seeing as". Creswell [32] further explains that "seeing as" in qualitative analysis involves recognizing recurring patterns, which aids in constructing meaning.

In our study, both "seeing" and "seeing as" were applied through multiple, iterative readings of the data to identify patterns and subsequently construct themes. This iterative process comprised several stages (outlined in Table 1): (a) data reduction (coding), (b) data categorization, (c) theme construction and presentation (thematization), and (d) summarization (drawing conclusions and verification). According to Ely [30], a theme is defined as a recurring pattern of expressions shared by multiple informants in response to a question, or as a significant statement reflecting strong emotional resonance.

In this study, participants' statements from the interviews (i.e., codes) were initially divided and organized into categories. Themes were then constructed within each category, encompassing participants' experiences, reactions, and attitudes toward the robots. This included both emotional responses and practical insights regarding the challenges and advantages encountered during interactions with the robots.

**Table 1.** Examples of the coding framework for the thematic analysis of the participants' interactions.

| Codes | Categories | Main Theme | Sub-Theme |
|---|---|---|---|
| "A robot should be able to do several things, it could be changed just by pressing buttons so it can do this task instead". | Development<br>Efficiency<br>Flexibility | Adaptability | |
| "[. . .] they need to be developed a bit more so that it feels like a human encounter, as it seems that many highly value this". | Quality<br>Speed<br>Development<br>Opportunities | Adaptability | Development |
| "Certainly, there are uses, tasks where there is no need for human contact but where things need to be fetched or information conveyed". | Usage Areas<br>No Need for Humanity | Adaptability | Usage Areas |
| "[. . .] Pepper was a bit slow to respond so sometimes it was unclear if my question had been received, so then I wanted to repeat the question [. . .]". "The other robot [Pepper] was a bit better at this [showing human emotions]. It seemed like it cheered up when it was told it had done a good job, so there was a bit of a difference with the other [Double], which was just a screen". | Pepper's Efficiency<br>Lack of Communication<br>Flow<br>Interaction<br>Interchangeability<br>Human Emotions | Interaction and Behavior | |
| "Yes, maybe that this little humanoid robot was, I thought it was fun when it waved and nodded and was supposed to be happy". | Behavior<br>Movements and Gestures<br>Conveying Emotions | Interaction and Behavior | Gestures and Body Language |
| "I was less honest with the robot [Pepper]. It felt like it would be sad if you said I can do it better myself. At least that was the feeling". | Human Emotions<br>Credibility<br>Honesty | Interaction and Behavior | Interaction |
| "[I] think that I really see and know that it is a robot, while people move more, and just act as they think, maybe ask more follow-up questions and such than what a robot does". | Autonomy<br>Movements and Gestures<br>Communication | Autonomy | |
| "Yes, but it feels a bit scary [future with AI] but it feels like something you might just have to accept". | Negative towards the future with AI<br>Scary | Autonomy | The Future |
| "I find it hard to see tasks that require trust and a lot of knowledge from the one you're getting help from, that robots will be able to replace there". | Trust and Confidence<br>Replacing Humans | Credibility | |
| "I think many feels that they are not so excited about dealing with robots because they don't feel like real people". | Lack of Human-likeness<br>Lack of Credibility | Credibility | Human-likeness |
| "[. . .] I think it's a disadvantage to remove that, the feeling of meeting an actual person". | Human Interaction<br>Replacing Humans | Credibility | Interchangeability |

The themes derived from the empirical data, which will be detailed in the Result section, were constructed through several iterative steps in the analysis process (see Table 1).

The final phase of analysis begins once all themes have been sufficiently developed to produce a cohesive report. During this phase, the researcher's role is to compile and organize the analysis into a coherent and readable paper that presents the final themes and highlights their relevance to the study [30].

The themes identified from the empirical data emerged through multiple iterative steps in the analysis process. In the following section, quotations are provided not as direct evidence, but as illustrative examples of the themes derived from the analysis. It is important to note that all quotations have been translated from Swedish into English for clarity.

### 2.8. Assessment of the Research Environment

The experimental environment was designed to simulate a warehouse scenario where participants could interact with the robots in a controlled and consistent setting. This environment aimed to minimize external variables while providing a practical context for evaluating user-robot interactions. The controlled nature of the environment offered several advantages. It ensured that all participants experienced identical conditions, allowing for a fair comparison of their interactions with Pepper and Double 3. This consistency facilitated the analysis of key variables, such as user perceptions of adaptability, autonomy, and emotional engagement. Additionally, the setting enabled the researchers to isolate specific design elements of the robots that influenced user responses.

However, the controlled environment also presented limitations. It may not fully replicate the complexity and unpredictability of real-world settings, where factors such as social dynamics, environmental noise, and multitasking demands could impact user interactions. Participants' behaviors and responses might differ in naturalistic environments, limiting the generalizability of findings to broader applications.

To address these limitations in future research, we recommend incorporating semi-controlled or real-world scenarios that more closely mimic the intended use cases of the robots. For example, integrating the robots into workplace or educational settings could provide richer data on how they function in dynamic, real-life contexts. Furthermore, gathering feedback from a more diverse participant group in these environments would enhance the robustness and applicability of the results.

Overall, the experimental environment provided valuable insights into user experiences with Pepper and Double 3, serving as a foundational step toward understanding human–robot interaction. By acknowledging its strengths and limitations, this study lays the groundwork for designing more comprehensive future research methodologies.

## 3. Experimental Environment and Results

The findings are organized into two primary areas: observations and interviews. The first area, observations, is categorized into three themes that capture participants' direct interactions with the robots, including their behaviors, reactions, and non-verbal cues. The second area, interviews, is structured around four main themes that explore participants' verbalized thoughts and experiences, providing deeper insights into their perceptions and emotional responses to the robots. Together, these themes offer a nuanced understanding of the complexities and dynamics of human–robot interaction within the context of this study.

### 3.1. Findings from the Observations

Functionality. During the tests, a general observation was that the Pepper robot operated more slowly than the Double 3 robot, which participants frequently commented on, often reflecting their reactions through body language. Although one participant noted that the Double 3 robot was also slow, none of the participants could walk at their normal pace due to the slower speeds of both robots. Occasionally, Pepper's connection issues further slowed its movements, causing it to veer off course near doorways or to travel too far, which extended the duration of certain tests.

Both robots encountered some minor technical issues. Pepper exhibited longer response times, which led participants to display puzzled expressions or, in some cases, laugh. Additional issues included start-up delays and difficulties in movement patterns, such as standing too close to shelves and struggling to wave. Technical disruptions like unexpected movements were also observed with Pepper. While some participants adapted well to these disruptions, others found them challenging, which impacted the overall testing process. In contrast, no major issues were reported with the Double 3 robot; it generally functioned as expected, with participants perceiving it as relatively stable.

Following the robots, especially Double 3, appeared relatively straightforward for participants. However, Pepper was perceived as slow and somewhat jerky in its movements. There were also misunderstandings regarding positioning, particularly with Double 3, as some participants initially thought they should walk in front rather than follow. This caused brief moments of confusion, though participants quickly adapted. Despite certain challenges with Pepper's speed and movement, most participants managed to keep pace. The results highlight, hence, notable differences in user perceptions of the Pepper and Double 3 robots, influenced by their respective designs and operational modes. While Pepper's humanoid embodiment aimed to foster emotional engagement, technical issues, including slower movement speeds and response delays, occasionally detracted from its performance. These challenges were recognized as factors potentially shifting focus away from embodiment to functionality.

It is also acknowledged that the Double 3 robot, optimized for teleoperation, had an advantage in delivering smoother and more predictable interactions due to its design. These disparities underline the importance of considering both embodiment and operational functionality when evaluating user experiences. Despite these differences, participants reported valuable insights into the unique strengths and weaknesses of each robot, contributing to the study's broader understanding of HRI.

Communication. The instructions provided by the robots were generally perceived as clear, with minimal need for repetition. However, some participants struggled to remember specific instructions and mentioned missing the human-like aspect of gestures to indicate item locations. In one instance, a participant decided to locate the item independently when the Pepper robot failed to respond for an extended period due to a malfunction.

When a participant attempted spontaneous questions with Pepper, communication was perceived as limited and inefficient due to Pepper's slower response times, which created confusion. Most participants, however, had little difficulty locating items after reaching their destinations. Some encountered minor issues with Double 3, likely due to the storage setup rather than any deficiency in the robot's instructions. A few participants struggled to locate items with Pepper, often due to its orientation, which sometimes led to misleading pointing gestures.

User Behavior, Experience, and Attitude. Participants' body language revealed signs of uncertainty or impatience, particularly when Pepper took longer to establish eye contact or respond. While this occurred to a lesser extent with Double 3, it was notably more common with Pepper due to its slower response times and frequent pauses in movement or speech. Despite these challenges, most participants remained patient and engaged throughout the tests.

Attitudes toward the robots varied. Some participants were consistently positive before, during, and after the test, while others expressed initial hesitation, especially toward Pepper, or grew skeptical due to technical issues encountered during the interactions. Several participants noted Pepper's slowness, which impacted their overall perception of the robots. One participant, although generally positive, maintained a cautious distance and

expressed a more skeptical view of increased robot usage in the future. Another participant felt uncomfortable with Pepper's almost human-like appearance and movements.

Overall, participants demonstrated patience during the tests. Most displayed a willingness to accommodate the robots' slower speeds, likely influenced by an understanding that the test was part of a research study. Minor signs of impatience, such as looking away, softly chuckling, or asking questions when Pepper took longer than expected, were observed. In general, participants managed the robots' slow responses well, though some signs of restlessness and curiosity about the robots' response times were evident.

### *3.2. Findings from the Interviews*

The results from the interviews are organized into five main themes: adaptability, usage areas, development, interaction and behavior, and credibility. Each theme includes sub-themes that provide additional insight into participants' perceptions and experiences with the robots.

Adaptability. This theme addresses the importance of a robot's flexibility to adapt to various situations, aligning with Mutlu's [7] concept of situativity—how a robot's behavior adapts to its environment, influencing user engagement and perceived realism. Participants expressed that robots should adapt to different user needs and preferences, with one envisioning that robots could perform different tasks with simple configuration adjustments, as exemplified by the statement, "A robot should be able to do several different things; you could change it just by pressing buttons so that it can do this task instead". These findings highlight the perceived importance of adaptability for enhancing the acceptance and accessibility of robot technology across multiple environments.

Usage Areas. Participants suggested that adaptable robots could serve a variety of purposes, from performing simple chores like cleaning to providing guidance at airports or customer service in healthcare. Examples included support in nursing homes, assistance with documentation in construction, and aiding new employees in companies. A common theme among participants was that robots could best handle tasks that do not require human interaction. One participant stated, "There are tasks where there is no need for human contact but where things need to be fetched or information conveyed", reflecting a general perception of robots as ideal for routine, low-interaction tasks.

Development. The development theme emphasizes participants' desire for robots to exhibit a more "human touch" in their interactions, a feature they felt would improve user experience. One participant suggested that for robots to be widely accepted, they should offer a more human-like experience, remarking, "[. . .] they need to be developed a bit more so that it feels like a human encounter, as it seems that many highly value this". Another participant highlighted technical improvements, suggesting that communication enhancements would aid interaction, particularly for users with hearing difficulties. This theme underscores the need for ongoing development in both technical performance and human-like qualities.

Interaction and Behavior. This theme explores participants' ease or challenges in interacting with the robots, referencing Mutlu's [7] ideas on interactivity and proxemics, which describe the influence of interaction quality and physical proximity on user comfort. Participants generally found the robots, especially Pepper, helpful, noting that Pepper displayed positive, welcoming behavior and responded well to feedback. One participant observed, "Yes, the other robot [Pepper] was better at that. It seemed like it perked up when told it had done a good job". However, some participants experienced confusion around where to position themselves relative to the robots, especially when Pepper was slow to respond, creating uncertainty. This feedback highlights how the robots' response times and movement speed could impact user experience.

Gestures and Body Language. Participants noted that Pepper's gestures and body language varied in perceived human likeness, with some finding its movements "cute" and others "uncomfortable". While some participants felt Pepper's expressive gestures made it more human-like, others found these gestures disconcerting. For instance, one participant remarked, "The white one [Pepper] was uncomfortable. . .too much hand, neck, and waist movement to be reasonable for a robot". This divergence in perception underscores the impact of gestures and body language on users' comfort levels.

Interaction. Participants had mixed responses to the robots' perceived emotional capabilities. Some expressed hesitation in providing negative feedback to Pepper, worrying it might "feel" hurt, while others found it easier to be honest with Pepper because they did not consider it sentient. One participant explained, "I felt less honest with Pepper. . . it felt like it would be sad if I said, 'I can do it better myself.'" This theme illustrates how perceptions of robot sentience can influence user behavior, reflecting a complex dynamic between empathy and detachment in human–robot interactions.

Autonomy. The autonomy theme concerns a robot's ability to operate independently, relating to Mutlu's [7] concept of agency, which involves a robot's perceived control and influence in its environment. Participants viewed human autonomy as superior, emphasizing that human spontaneity and the ability to ask follow-up questions create a richer interaction experience. As one participant stated, "I really see and know that it is a robot, while people move more and just act as they think". This perspective suggests that limited robot autonomy affects interaction quality, with participants expressing a preference for human-like responsiveness.

The Future. Participants expressed both interest and concern about the future of robots and AI, especially in terms of increasing autonomy. Some were wary of robots gaining "consciousness" and acting independently, with one participant stating, "One is always afraid that the robot might. . . somehow gain its own consciousness". These reflections suggest a mixture of fascination and apprehension about future AI developments, highlighting the need for careful planning to manage potential societal impacts.

Credibility. This theme captures participants' skepticism regarding robots' ability to handle tasks requiring trust and expertise, aligning with Mutlu's concept of believability [7]. Participants doubted the robots' capabilities in complex or knowledge-intensive roles, with one expressing, "I find it hard to see tasks that require trust and perhaps a lot of knowledge. . . that robots will be able to replace there". This indicates an ongoing trust barrier for robots, especially in scenarios requiring high competence and credibility.

Human likeness. The human-likeness theme focuses on participants' varied responses to the robots' human-like qualities. While some participants appreciated these qualities, finding them engaging, others found them unsettling. A participant noted, "The other robot [Pepper] was much more uncomfortable. . .it moved half like a human". This theme suggests that while human likeness can enhance engagement for some users, it may lead to discomfort for others, highlighting the importance of balancing human-like features to suit diverse user preferences.

Interchangeability. Finally, interchangeability addresses participants' views on the extent to which robots could replace humans. While they acknowledged robots' potential to perform simple, routine tasks, participants expressed hesitation about fully replacing human roles in contexts requiring empathy and trust, such as healthcare. One participant remarked, "If I'm at a hospital and need help. . . I prefer it to be a human and not a robot". This sentiment underscores the current limitations of robots in high-stakes or human-centric roles. However, some participants acknowledged that with advancements in speed and communication, robots could eventually replace humans in specific roles: "If they become faster and more advanced, they could get top marks for tasks like these".

## 4. Discussion and Conclusions

This study aimed to explore and understand user experiences and reactions during interactions with two distinct types of robots, Pepper and Double 3. The central research question focused on how these interactions influence participants' perceptions, attitudes, and responses toward these AI embodiments.

The study findings reveal diverse participant reactions to each robot. Participants expressed a desire for robots that could be easily adjusted to perform various tasks, underscoring the importance of flexibility in robot design (i.e., adaptability). Participants had mixed responses to the robots' imitation of human emotions, with some finding it uncomfortable while others were more receptive (i.e., interaction and behavior). Human interactions were perceived as more nuanced and autonomous compared to the robots, highlighting limitations in the robots' autonomous capabilities (i.e., autonomy). Additionally, participants were hesitant to trust the robots in complex scenarios, emphasizing the need for reliability and credibility in AI systems.

The theme of adaptability aligns with Mutlu's [7] concept of situativity, where a robot's behavior is adapted to its environment, enhancing user engagement and realism. This desire for adaptable robots underscores the importance of user-centered design principles in integrating AI into everyday life.

The mixed reactions to the robots' human-like emotional displays connect with Mutlu's [7] mechanisms of interactivity and proxemics. The variation in user comfort levels and perceptions of social presence reinforce the complexity of designing AI systems that balance human-like behavior with mechanical functionality.

Perceived limitations in the robots' autonomy reflect ongoing challenges in AI, echoing Mutlu's [7] exploration of agency. Although Pepper and Double 3 represent advancements in AI, participants noted their lack of nuanced, independent actions characteristic of human interaction, underscoring the need for further development in autonomous capabilities.

Participants' hesitation to trust the robots in complex tasks resonates with the concept of believability [7], suggesting that trust and perceived competence are vital for user acceptance of AI technologies. Discomfort with Pepper's human-like traits also reflects the challenges of navigating the "uncanny valley" in AI design, where overly human-like behavior can be unsettling, underscoring the importance of empathetic yet non-intrusive AI.

This study's findings regarding embodiment align with research by Cassell et al. [4] and Mutlu [7], who emphasize that the physical form of an AI agent significantly affects user perception and engagement. Participants responded differently to the humanoid form of Pepper and the utilitarian design of Double 3, influencing their interaction and comfort levels. This distinction supports Mutlu's observations that embodiment impacts user comfort and ease of interaction with AI agents.

Consistent with results presented by D'Alfonso et al. [8], some participants felt more comfortable sharing thoughts with Pepper, likely due to its non-threatening, humanoid appearance. However, while it was obvious that Double 3 was controlled by a human, the mode of control of the Pepper robot was more unclear. This may also have impacted the participants' comfort.

The participants' perspectives on using robots like Pepper and Double 3 across various sectors—such as healthcare, education, and remote work—mirror insights from Sheridan [6]. The results suggest that specialized robots can enhance user experience and efficiency in these areas, suggesting that robots designed for specific tasks offer practical value. Moreover, the impact of humanoid features on human–robot interactions, such as communication and relationship-building, aligns with Breazeal [20] and Wirtz's [17] views on the transformative potential of social robots. Participants' responses to Pepper's expressive features and Double 3's functional design demonstrate the potential for robots

to positively influence customer service and remote collaboration, echoing observations by Yang et al. [21] and Lee and Takayama [23].

### 4.1. Conclusions

In conclusion, both Pepper and Double 3 are perceived as user-friendly. Participants appreciated Pepper's intuitive interaction and Double 3's practical utility, suggesting that these robots bridge gaps between virtual and physical interactions. The results also highlight the importance of designing robots that are adaptable, interactive, autonomous, and credible, and offer guidance for future development. This study underscores the potential of human–robot interaction to transform various domains, including healthcare, education, and remote collaboration. By addressing user needs for adaptability, autonomy, and reliability, robots can significantly enhance user experiences and operational efficiency. However, the findings also highlight critical challenges, such as achieving a balance between human-like traits and functional usability, which remain pivotal for advancing robotic design.

Furthermore, the study emphasizes the importance of user-centered design in robotics. Participants' feedback on both robots illustrates the need for iterative development processes that prioritize user comfort, trust, and engagement. Designing robots that can effectively navigate complex social and professional contexts is essential for broader acceptance and integration of AI technologies.

### 4.2. Design Recommendations for Robot Developers

Based on the findings of this study, several explicit design recommendations can be made for robot developers. First, robots should incorporate synchronized gestures and facial expressions to create more natural and engaging interactions, as these features were particularly effective in fostering trust and emotional engagement, especially with the Pepper robot. Additionally, robot behaviors should be tailored to specific tasks and user needs. For instance, the Double 3 robot's telepresence capabilities could be optimized by improving its navigation interface to enable smoother transitions in confined spaces.

Developers should also carefully consider the physical appearance of robots in relation to their intended roles, as humanoid features were found to increase perceived usability in social contexts, as demonstrated by the Pepper robot. Furthermore, incorporating real-time feedback mechanisms during interactions can significantly enhance usability and reduce confusion. This was evident in the positive reception to both robots' verbal guidance during the simulated warehouse tasks. Finally, developers should design robot systems for scalability, ensuring that they can be adapted for broader applications without extensive reprogramming. This is particularly important for versatile robots like the Double 3, which can serve multiple functions across domains. By implementing these recommendations, developers can create robots that not only meet functional requirements but also enhance user satisfaction and trust, ultimately contributing to the broader acceptance and integration of robots in diverse settings.

Looking ahead, this study offers a foundation for future research to explore long-term interactions, diverse user groups, and the ethical implications of robotics. By addressing these dimensions, the field can move closer to developing robots that are not only effective tools but also empathetic companions in human environments.

### 4.3. Limitations and Future Research

This study involved a small group of participants, limiting the diversity of perspectives and experiences observed. The specific focus on the Pepper and Double 3 robots may affect the generalizability of findings, as different robot models with varying capabilities could elicit different user responses. Additionally, while the study provided insights into initial

reactions, it did not capture the dynamics of prolonged robot interaction. The findings are also influenced by the technological design of the robots, which shaped participants' experiences. Conducting the study in a controlled environment may not fully reflect real-world settings, where user needs and responses are more complex.

This study adopted a qualitative approach to provide an in-depth understanding of user experiences during human–robot interactions. While a quantitative approach could offer measurable findings, the focus here was on capturing nuanced perceptions to inform future designs. This reflects the early stage of research in this area, where foundational insights are crucial before scaling to larger, quantitative studies. While the study focused on a single interaction scenario within a simulated warehouse, this limitation is acknowledged. Future research could include multiple scenarios to broaden understanding of how robot embodiment influences user interactions in varied contexts, enhancing the robustness of findings. Despite these limitations, the study offers valuable insights into user experience and provides practical design implications. It also enhances understanding of ethical and social considerations in AI and robotics, serving as a methodological model for HRI studies and informing policy development.

Future research should focus on enhancing key features of AI systems and robots, such as adaptability, interactivity, and autonomy. Addressing limitations like slow movement speeds and response delays in robots like Pepper and Double 3 is essential. Improved simulation of human behaviors and gestures could further enhance user comfort and acceptance of AI technology. Enhanced communication features and reduced response times should remain priorities for optimizing user interactions. Promoting awareness of robots' limitations and capabilities can foster more effective communication, improving overall interaction quality. Exploring the ethical and societal implications of integrating robots into daily life—such as privacy, job displacement, and human–robot relationship dynamics—will provide critical insights as society moves toward increased AI integration.

**Author Contributions:** Conceptualization, E.M.-O., S.B. and T.H.; methodology, E.M.-O.; software, S.B. and T.H.; validation, E.M.-O.; formal analysis, E.M.-O.; investigation, E.M.-O., S.B. and T.H.; resources, S.B. and T.H.; data curation, H.A., A.H., M.L. and S.W.; writing—original draft preparation, H.A., A.H., M.L. and S.W.; writing—E.M.-O.; visualization, H.A., A.H., M.L. and S.W.; supervision, E.M.-O., S.B. and T.H.; project administration, E.M.-O. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research received no external funding.

**Institutional Review Board Statement:** Ethical review and approval were waived for this study due to the observational nature of the study.

**Informed Consent Statement:** Informed consent was obtained from all subjects involved in the study.

**Data Availability Statement:** Data are not publicly available due to privacy restrictions. However, data may be available from the authors upon reasonable request.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## References

1. Zhu, Y.; Wang, T.; Wang, C.; Quan, W.; Tang, M. Complexity-Driven Trust Dynamics in Human–Robot Interactions: Insights from AI-Enhanced Collaborative Engagements. *Appl. Sci.* **2023**, *13*, 12989. [CrossRef]
2. Rossi, A.; Holthaus, P.; Moros, S.; Lakatos, G.; Andriella, A.; Scheunemann, M.; Van Maris, A. Trust, Acceptance and Social Cues in Human–Robot Interaction (SCRITA). *Int. J. Soc. Robot.* **2024**, *16*, 1047–1048. [CrossRef]
3. Sorrentino, A.; Fiorini, L.; Cavallo, F. From the Definition to the Automatic Assessment of Engagement in Human–Robot Interaction: A Systematic Review. *Int. J. Soc. Robot.* **2024**, *16*, 1641–1663. [CrossRef]
4. Cassell, J.; Sullivan, J.; Prevost, S.; Churchill, E. *Embodied Conversational Agents*; MIT Press: Cambridge, MA, USA, 2000.

5. Carreno-Medrano, P.; Tian, L.; Allen, A.; Sumartojo, S.; Mintrom, M.; Coronado, E.; Venture, G.; Croft, E.; Kulic, D. Aligning Robot's Behaviors and User's Perception Trough Participatory Prototyping. *arXiv* **2021**, arXiv:2101.03660v1. [CrossRef]
6. Sheridan, T.B. Human–Robot Interaction: Status and Challenges. *Hum. Factors* **2016**, *58*, 525–532. [CrossRef]
7. Mutlu, B. The Virtual and the Physical: Two Frames of Mind. *iScience* **2020**, *24*, 101–117.
8. D'Alfonso, S.; Santesteban-Echarri, O.; Rice, S.; Wadley, G.; Alvarez-Jimenez, M. Artificial Intelligence-Assisted Online Social Therapy for Youth Mental Health. *Front. Psychol.* **2017**, *8*, 796. [CrossRef]
9. Chen, S.; Smith, R.; Jones, D. Advances in Robotic Assistants for Elderly Care. *J. Healthc. Robot.* **2022**, *15*, 123–135.
10. Ramirez, L.; Carter, B. The Role of Social Robots in Enhancing Customer Engagement in Retail. *AI Soc.* **2023**, *38*, 89–104. [CrossRef]
11. Kim, J.; Lee, H. Social Robots in STEM Education: An Empirical Study. *Educ. Technol. Soc.* **2023**, *26*, 50–67.
12. Su, H.; Qi, W.; Chen, J.; Yang, C.; Sandoval, J.; Laribi, M.A. Recent Advancements in Multimodal Human–Robot Interaction. *Front. Neurorobot.* **2023**, *17*, 1084000. [CrossRef] [PubMed]
13. Pandey, A.K.; Gelin, R. A Mass-Produced Sociable Humanoid Robot: Pepper: The First Machine of Its Kind. *IEEE Robot. Autom. Mag.* **2018**, *25*, 40–48. [CrossRef]
14. Tanaka, F.; Matsuzoe, S. Children Teach a Care-Receiving Robot to Promote Their Learning: Field Experiments in a Classroom for Vocabulary Learning. *J. Hum.-Robot Interact.* **2012**, *1*, 78–95. [CrossRef]
15. Zhao, Y.; Okamoto, T. The Effects of Telepresence Robot on Communication in a Simulated Classroom. *Int. J. Soc. Robot.* **2015**, *7*, 661–671.
16. Neustaedter, C.; Venolia, G.; Procyk, J.; Hawkins, D. To Beam or Not to Beam: A Study of Remote Telepresence Attendance at an Academic Conference. *Hum.-Comput. Interact.* **2016**, *31*, 45–102.
17. Wirtz, J.; Patterson, P.G.; Kunz, W.H.; Gruber, T.; Lu, V.N.; Paluch, S.; Martins, A. Brave New World: Service Robots in the Frontline. *J. Serv. Manag.* **2018**, *29*, 907–931. [CrossRef]
18. Siciliano, B.; Khatib, O. *Springer Handbook of Robotics*, 2nd ed.; Springer: Berlin/Heidelberg, Germany, 2016.
19. Double Robotics. *Double 3: A New Era of Telepresence*; White Paper; Double Robotics, Inc.: Redwood City, CA, USA, 2019.
20. Breazeal, C. Social Interactions in HRI: The Robot View. *IEEE Trans. Syst. Man Cybern. C* **2004**, *34*, 181–186. [CrossRef]
21. Yang, G.Z.; Bellingham, J.; Dupont, P.E.; Fischer, P.; Floridi, L.; Full, R.; Wood, R. The Grand Challenges of Science Robotics. *Sci. Robot.* **2018**, *3*, eaar7650. [CrossRef]
22. Benitti, F.B.V. Exploring the Educational Potential of Robotics in Schools: A Systematic Review. *Comput. Educ.* **2012**, *58*, 978–988. [CrossRef]
23. Lee, M.K.; Takayama, L. "Now, I Have a Body": Uses and Social Norms for Mobile Remote Presence in the Workplace. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, Vancouver, BC, Canada, 7–12 May 2011.
24. Aldebaran. Pepper the Humanoid and Programmable Robot. Available online: https://www.aldebaran.com/en/pepper (accessed on 19 September 2023).
25. Alemi, M.; Meghdari, A.; Ghazisaedy, M. The Impact of Social Robotics on L2 Learners' Anxiety and Attitude in English Vocabulary Acquisition. *Int. J. Soc. Robot.* **2015**, *7*, 523–535. [CrossRef]
26. Smith, A.C. Exploring the Efficacy of Telepresence Robots in Health Care Settings. *J. Telemed. Telecare* **2017**, *23*, 439–446.
27. Rietz, F.; Sutherland, A.; Bensch, S.; Wermter, S.; Hellström, T. WoZ4U: An Open-Source Wizard-of-Oz Interface for Easy, Efficient and Robust HRI Experiments. *Front. Robot. AI* **2021**, *8*, 668057. [CrossRef] [PubMed]
28. Design-Based Research Collective. Design-Based Research: An Emerging Paradigm for Educational Inquiry. *Educ. Res.* **2003**, *32*, 5–8. [CrossRef]
29. Wang, F.; Hannafin, M.J. Design-Based Research and Technology-Enhanced Learning Environments. *Educ. Technol. Res. Dev.* **2005**, *53*, 5–23. [CrossRef]
30. Ely, M. *Doing Qualitative Research*; Falmer Press: London, UK, 1991.
31. Boyatzis, R. *Thematic Analysis and Code Development: Transforming Qualitative Information*; Sage: Thousand Oaks, CA, USA, 1998.
32. Creswell, J.W. *Qualitative Inquiry and Research Design: Choosing Among Five Approaches*, 3rd ed.; Sage: Thousand Oaks, CA, USA, 2013.


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